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Design of a Multiple Phased Convolution Model for Pre-emption of Breast Cancer via Extensive Learning Operations (MPC2EL)

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Abstract

Breast cancer diagnosis from mammographic scans is a complex challenge, demanding accurate and efficient methodologies. Existing approaches often rely on time-consuming manual feature extraction or lack scalability when employed with diverse clinical image sets. To address these limitations, this paper presents an innovative solution: the MPC2EL (Multiple Phased Convolution model for Pre-emption of Breast Cancer via Extensive Learning Operations). The core problem tackled by this research is the need for improved breast cancer detection, especially concerning scalability, accuracy, and practicality. To achieve this, MPC2EL leverages extensive learning operations and multiple phased convolutions to process extensive clinical datasets containing both cancerous and benign images. Our novel approach begins with the conversion of these images into multimodal feature sets, encompassing frequency, cosine, wavelet, and Gabor domains. These feature sets enable enhanced classification of images into different cancer stages. The proposed methodology lies in a Dual 1D Convolutional Neural Network (D1D CNN) architecture, capable of categorizing images initially into malignant and benign types and subsequently refining classification for malignant images. Experimental results reveal a substantial 5.9% increase in classification accuracy and a 4.5% improvement in recall compared to conventional deep learning methods. Moreover, the model demonstrates lower processing delay and higher precision, making it a promising candidate for real-world clinical deployment.

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Keywords: *Cancer; Mammogram; Multiple; Phased; CNN; Gabor; Cosine.*

1. Introduction

The change in lifestyles of individuals throughout the globe has led to an increase in the higher number of cancer cases identified at a younger age. One of the most prominent health risks for women is breast cancer. Breast cancer has the greatest rates of morbidity and mortality among all diseases that affect women [1]. In the year 2020, little more than 2,3 million women had a breast cancer diagnosis, and 0.68 million women died from the disease. A significant proportion of breast cancer deaths may be linked to late disease detection. Cancer may be detected early through imaging techniques such as X-rays, ultrasound, thermography, diagnostic mammography,

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and histology, sometimes called biopsies [2][3]. During the first phases of a patient's screening procedure, mammography is often used. This is because of its low price and excellent sensitivity to even the tiniest irregularities. It is now believed that a biopsy is the most effective way for obtaining a definite diagnosis. After the sample has been preprocessed and separated from the biopsy material, pathologists examine it under a microscope to ascertain the results. The findings of this examination are highly dependent on the pathologist's degree of knowledge. The use of computer-assisted diagnosis (CAD), whose findings are much more accurate and reduce the stress and burden of pathologists, is crucial.

This paper's contributions lie in its innovative feature representation, the introduction of the D1D CNN architecture, significant improvements in breast cancer classification performance, and its practicality for clinical use, collectively advancing the field of breast cancer detection and diagnosis.

1.2 Research Goals and Aims

The primary research goals and aims of this study are as follows:

I. **Enhancing Breast Cancer Detection Accuracy**: The MPC2EL model employs multiple phased convolutions and extensive learning operations to enhance the accuracy of breast cancer detection from mammographic scans. The research will provide reliable breast cancer diagnosis by providing a robust and efficient tool for early detection.

II. Scalability to Clinical Image Sets: Addressing the challenge of scalability, this study aims to create a model that can effectively handle clinical image sets, which often exhibit variations in terms of image quality, diversity of cases, and data volume. The goal is to ensure that the MPC2EL model can be applied seamlessly to real-world clinical scenarios.

III. Multimodal Feature Representation: This research aims to introduce a novel approach by converting histopathological images into multimodal feature sets, encompassing frequency, cosine, wavelet, and Gabor domains. The use of diverse feature representations will increase the accuracy of breast cancer classification and to uncover potentially subtle indicators of malignancy.

IV. Dual 1D Convolutional Neural Network (D1D CNN): A significant aim of this study is to design and implement a D1D CNN architecture within the MPC2EL model. This neural network serves the purpose of categorizing feature sets into malignant and benign types initially, and subsequently classifying malignant images into different cancer stages. The research seeks to demonstrate the effectiveness of this dual CNN approach.

V. Performance Improvements: The goal is to showcase tangible improvements in breast cancer classification performance. This includes an increase in classification accuracy, recall, and precision when compared to standard deep learning methods. The research aims to establish the practicality and effectiveness of the MPC2EL model for clinical deployment.

This paper's contributions lie in its innovative feature representation, the introduction of the D1D CNN architecture, significant improvements in breast cancer classification performance, and its practicality for clinical use, collectively advancing the detection and diagnosis of breast cancer.

2. Literature Review

The biopsy procedure is laborious, time-consuming, and heavily impacted by the pathologist's expertise, and hence may lead to erroneous conclusion formation, it is crucial to use CAD. CAD findings are also much more precise [5][6]. Computer-aided diagnostics, on the other hand, enables more precise issue resolution. This is due to consideration of the complexity of the tumor's shape and growth, as well as other minor characteristics at the fragile early stage of the disease.

The computer-aided diagnosis (CAD) process begins with locating the precise area identified in the preprocessed mammogram, then proceeds to extract features of the tumor based on the treating physician's expert knowledge, and then concludes with the diagnosis of benign or malignant tumors based on vectors such as shape, morphology, density, and texture [3][4]. Deep learning is a very effective method for gaining knowledge of features, semantic rules, and other crucial information directly from data. It is a data-driven approach that follows a data-driven methodology. Deep learning algorithms were developed to reduce the number of false-negative interpretations that may arise during the screening phase. These interpretations may be brought on by the existence of complex vectors, subtle outcomes, or radiologists' distractions. Recent studies have shown that computer-aided design (CAD) and medical image processing, which utilizes machine learning and deep learning, produce more accurate results than the pathologist's diagnosis [3-7]. CNN is a component of deep learning which is used to detect brain tumors in MRI scans. These techniques have been used to develop automated methods that can accurately detect and diagnose breast cancer. This technology has the potential to revolutionize the way breast cancer is detected and treated. [7-10]. Histopathological images are typically classified using small datasets in most studies. The reason behind this is the lack of availability of large datasets and the computational challenges associated with handling large image databases. In an attempt to identify a solution, a collection of breast cancer histological images known as BreaKHis was published recently [11]. After finishing the experiment, the researchers discovered that the baseline performance for categorizing tumors as benign or malignant had an accuracy rate of 80–85%. To retrieve features from histopathology images during the examination, multiple deep learning approaches were used. In contrast, the traditional approaches used in the past needed a high degree of professional expertise and relevant data.

Consequently, the success rate of models that employ deep learning to learn from hierarchical representations is high. The most major advantage is that these models may effectively extract and organize a number of features without prior knowledge of the domain being represented. Deep learning strategies are generally binary, and they have been proposed for predicting the tumor subtype [12]. Nevertheless, several of these methods use multivariable classification [13]. Experiments have shown that predesigned networks like as AlexNet, MobileNet, and Inception may also be used [14]. Bayramoglu [18] applied the BreakHis dataset to develop a CNN model for predicting malignancy and image magnification. They observed that the accuracy of their model ranged from 80.6% to 83.3%. Gupta and Bhavsar [17] classified images using an integrated model of joint color-texture features and classifiers at different optical magnification levels. The recommended model produces an average accuracy of 87.53 percent, with the maximum degree of accuracy seen at 200 times magnification. Bardou et al. [16] used the BreakHis dataset in a number of methods in their analysis. The original technique entailed the development of two models for the extraction of handcrafted characteristics. SVM-trained models with Locality Constrained Linear Coding and Bag-of-Words were able to produce binary classifications with an accuracy of 96.15 to 98.33%. In the other method, CNN's accuracy was increased by using data augmentation techniques, along with "Handcrafted features +CNN" and "CNN Features + Classifier." De Matos et al. [20] used SVM-based classifiers to sift through the pre-trained dataset using CNN in order to identify areas of malignancy. Utilizing pre-processing techniques and training the model with pre-trained datasets may increase the accuracy of the model. Gupta and Chawla [15] conducted image classification using CNN in conjunction with SVM and Logical regression on a classifier that was trained using a previously collected dataset. According to the investigation's findings, CNN+LR provides more precise classifications than CNN+SVM. These results suggest that this is the case.

D. Li, L. Zhange et al. [21]. investigates the selection of effective feature descriptors for the classification of mammograms. By identifying and utilizing appropriate feature descriptors, it enhances the accuracy and reliability of mammogram-based breast cancer classification. H. Du, et al. [22] introduced the automatic classification of calcification morphology and distribution in breast mammograms. This innovative technique utilizes deep learning and graph convolutional neural networks to extract valuable insights from mammograms, ultimately contributing to a more comprehensive understanding of breast cancer characteristics and aiding in diagnosis

These studies show that current models for evaluating breast cancer exhibit either low scalability or high complexity when applied to clinical image sets. Therefore, further research is needed to develop more efficient and simpler models that can more accurately detect breast cancer. Limitations of existing models is shown in table 1.

Table 1. Limitations of existing models

Sr. No.	Limitation	Description
1	Limited Scalability	Some models lack scalability when applied to diverse clinical image sets, hindering real-world deployment.
2	Limited Large Datasets	Limited availability of large datasets hinders the development of robust breast cancer detection models.
3	Computational Challenges	Handling large image databases poses computational challenges in breast cancer image analysis.
4	Need for Data Augmentation	Data augmentation techniques are needed to improve CNN accuracy, indicating data scarcity issues.
5	Binary Classification	Many deep learning models focus on binary classification, which may not capture tumor subtypes.
6	Accuracy Variability	Model accuracy varies between studies, ranging from 80.6% to 98.33%, indicating inconsistencies.
7	Model Combination	Multiple models, such as CNN + SVM or CNN + Logical Regression, are used, suggesting a lack of consensus on the best approach.

3. Methodology

3.1 MPC2EL Architecture

MPC2EL model uses a Dual 1D Convolutional Neural Network (D1D CNN). This architecture is designed to efficiently process and classify the multimodal feature sets extracted from histopathological images.

i. **Input Layer**: The input layer receives the multimodal feature sets representing the histopathological images in various domains, including frequency, cosine, wavelet, and Gabor.

ii. **Convolutional Layers**: The network comprises multiple convolutional layers responsible for feature extraction and representation learning. Each convolutional layer is equipped with a set of learnable filters to detect specific patterns and features from the input data.

iii. **Pooling Layers:** In order to reduce dimensionality and preserve essential information, max-pooling layers are applied after each convolutional layer.

iv. **Flattening Layer**: The feature maps are flattened into a one-dimensional vector after the convolutional and pooling layers, preparing the data for dense layers.

v. **Dense Layers:** The flattened features are fed into dense layers, which perform classification tasks. These layers consist of several neurons which are responsible for learning complex relationships between features.

vi. **Output Layer:** The input images are classified into malignant or benign classes in output layer. In the case of malignant images, the dual CNN process is further employed for fine-grained classification into different cancer stages.

3.2 Experimental Setup

The experiments conducted to evaluate the MPC2EL model are crucial for assessing its performance and efficacy in breast cancer detection. The following details describe the experimental setup:

The experimental evaluation of the MPC2EL model was conducted using a comprehensive dataset of histopathology images obtained from https://www.kaggle.com/paultimothymooney/breast-histopathology-images. The dataset consists of a total of 2,77,524 images.

Preprocessing Prior to training and evaluation, the histopathology images underwent a series of preprocessing steps to ensure consistency and enhance their suitability for deep learning analysis. These steps included:

i. Resizing: All images were resized to a standardized resolution of 224x224 pixels to facilitate processing and reduce computational complexity.
ii. Normalization: Intensity values of the images were normalized to the range [0, 1] to improve convergence during training.
iii. Data Augmentation: To augment the dataset and enhance model robustness, random rotations, flips, and slight translations were applied to the images.

Model Architecture The MPC2EL model was implemented using the PyTorch deep learning framework. The architecture consists of the following components:

i. **Multimodal Feature Extraction**: Images were transformed into multimodal feature sets comprising frequency, cosine, wavelet, and Gabor domains. Sample parameters for these transformations include:
    - Frequency domain: Window size = 128, Overlapping factor = 0.5
    - Cosine domain: Number of coefficients = 32
    - Wavelet domain: Wavelet type = 'Daubechies', Decomposition levels = 4
    - Gabor domain: Number of scales = 5, Number of orientations = 8
ii. **Dual 1D Convolutional Neural Network (D1D CNN):** The core of the model, consisting of two parallel 1D CNNs, was designed to classify images into benign and malignant categories. Sample hyperparameters for the D1D CNN architecture include:
    - Number of convolutional layers: 4
    - Number of filters per layer: [64, 128, 256, 512]
    - Filter size: [3, 3, 3, 3]
    - Activation function: ReLU
    - Batch normalization: Applied after each convolutional layer
    - Max-pooling: After every two convolutional layers
    - Dropout rate: 0.3

Evaluation Metrics The performance of the MPC2EL model was assessed using the following evaluation metrics:

1. Accuracy: To measure the overall classification accuracy.
2. Recall: Specifically, recall for malignant cases, as early detection of breast cancer is a primary concern.
3. Precision: To measure the precision of classifying images as malignant.

Hardware and Software The experiments were conducted on a high-performance computing cluster with an NVIDIA Tesla V100 GPU. The software stack included Python 3.7, PyTorch 1.6, and various deep learning libraries for image processing and model evaluation process.

3.3 Data Preprocessing

After applying image enhancement techniques to the dataset, multiple layers of CNN are used to extract features from histopathology images and categorize the images based on the obtained features. The detailed procedure of dataset improvisation is depicted in figure 1, where database related operations consist of image resizing, removing biasedness from the dataset, and setting up a data generator to add more characteristics to the images such as rotation, horizontal flip, skewness, vertical flip, height shift, and width shift.

Pre-processing of data magnifies the specific features of images, required for study and also suppress the noises to improvise the performance with high efficiency. The first pre-processing approach deduce the grey value vacillations in the images and their successive removal from the original image. The wiener filter was pragmatic in a 3x3 neighborhood. The median value of the view is figured and detracted from the original image, finally concluding the final image after all abstractions.

The second pre-processing method gloomily the light grey cells in the image and progress sequentially. Initially, the image opening process is employed to scrutinize the image background model, followed with removal of that image from the original image. A 2D median filter is used to eliminate visual imbalances as part of the final phase, which also involves the elimination of noises like H&E paint imbalances. The salt and pepper noise were removed by applying Gaussian blur, Weiner filter was applied to remove motion blur and speckles and finally, adaptive histogram equalization was applied.

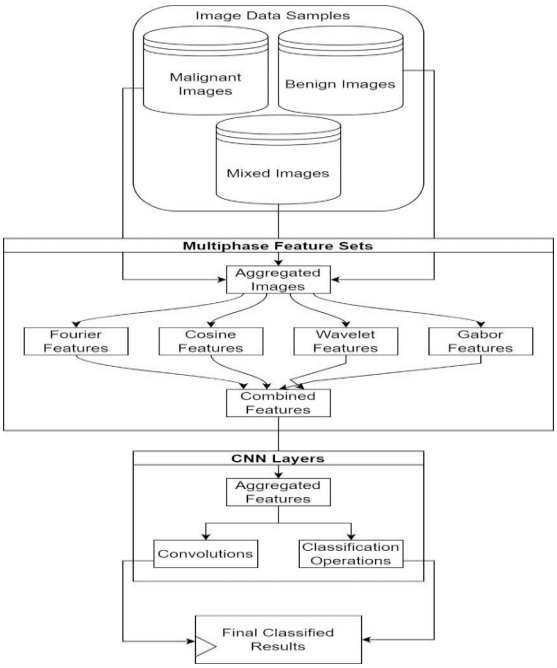


Fig.1. Steps in classification process

Over-preprocessing removes all information about the background texture of an image. The image is reduced to only cells and its boundaries. This is not useful for automatic feature extraction methods. The third step involves applying an adaptive threshold to the first step and then using a median filter with a 3x3 neighbourhood to remove smaller regions. The application of pre-processing on the histopathology images in a much more calculated manner improves the working of algorithms and improves model performance as compared to applying over-processing to the images.

3.4 Data Augmentation

The augmentation techniques used play a significant role in expanding the cardinality of the dataset since the performance of deep learning models is dependent extensively on the size of the training dataset selected. The experiment involved the inclusion of Fine-tuning 1 cycle. The CLR (Calcitonin Receptor) has a specific case known as the 1cycle learning rate in which just one cycle is employed, followed by cosine annealing to reduce the LR in the subsequent iterations. Super-convergence, when the test loss and accuracy swiftly approach optimal values and then remain constant, is attained using the 1cycle policy. The CyclicLR callback code, which has been modified for this study to include cosine annealing in the last rounds, is also used to implement this strategy.[22]

3.5 Model Development

For data classification, the softmax activation function has been used. The main goal of the proposed approach is to enhance the database after data augmentation to optimize the deep neural network's performance for categorizing histopathology images. The CNN model is trained on 72000 images and 18000 images were utilized for testing purposes. To improve the model accuracy and reduce the overall loss, adam optimizer was applied and categorical cross-entropy was applied as a loss function. 30 training epochs are used to train the model. The graphical steps of the CNN model applied for data classification are shown in figure 1 and the detailed architecture of the developed model is presented in figure 2

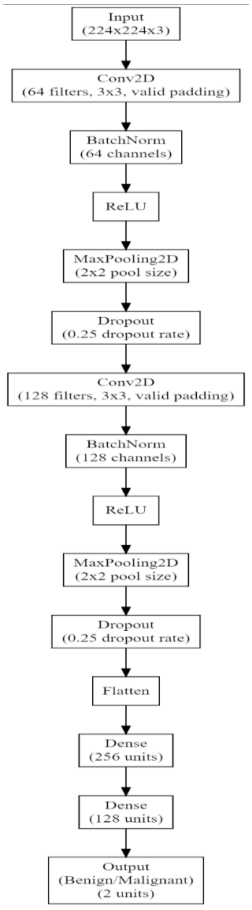


Fig.2. Architecture of the proposed model to identify cancer types

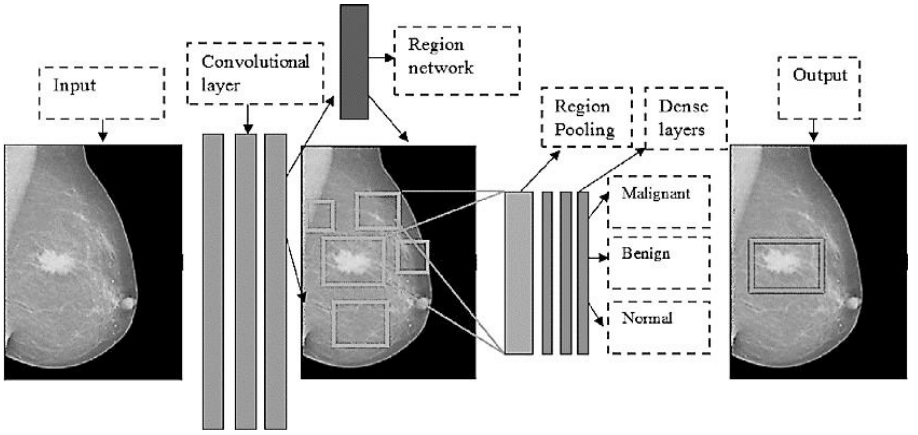


Fig.3. Procedure of data classification using CNN model

Figure 3 show how input images are first processed using a multiple-phased feature extraction model. This model extracts Fourier features using equation 1, Cosine features using equation 2, Gabor features using equations 3 and 4, and Wavelet features using equations 5 and 6. This helps to represent the images into different domain types.

$$DFT_i = \sum_{j=1}^{N_f} x_j * \left[cos\left(\frac{2*\pi*i*j}{N_f}\right) - \sqrt{-1} * sin\left(\frac{2*\pi*i*j}{N_f}\right)\right] \ldots \quad (1)$$

Where, x represents input report sample values, where N_f represents total number of features in the reports. Similar calculations are done for DCT sets,

$$F(DCT_i) = \frac{1}{\sqrt{2*N_f}} * x_i \sum_{j=1}^{N_f} x_j * \cos\left[\frac{\sqrt{-1}*(2*i+1)*\pi}{2*N_f}\right] \ldots \quad (2)$$

Similarly, the Gabor features are extracted via equation 3 as follows,

$$G(x,y)_s = e^{\frac{-x`^2+\partial^2*y'^2}{2*\emptyset^2}} * \cos\left(2*\frac{pi}{\lambda}*x'\right) \ldots \quad (3)$$

Where, x & y represents Cartesian components, and are evaluated as per equation 4,

$$x' = x*\cos(\phi) + y*\sin(\phi)$$
$$y' = -x*\sin(\phi) + y*\cos(\phi) \ldots \quad (4)$$

In this case, $\phi \in (0,\pi)$, and changes linearly in steps of $\pi/180$ to determine different components. These features are augmented via Wavelet components which are estimated as per equations 5 & 6,

$$W_a = \frac{x_i + x_{i+1}}{2} \ldots (5)$$
$$W_d = \frac{x_i - x_{i+1}}{2} \ldots (6)$$

These aggregated feature sets are combined, and given to the CNN for identification of tumor types. The CNN model extracts a wide variety of convolutional features via equation 7 & 8,

$$Conv_{out_i} = \sum_{a=-\frac{m}{2}}^{\frac{m}{2}} x(i-a) * LReLU\left(\frac{m+2a}{2}\right) \ldots \quad (7)$$

Where, m, a represents size of convolutional windows, and strides, while $LReLU$ represents a Leaky Rectilinear Unit, which is used for activation of features, and is represented via equation 8,

$$LReLU(x) = l_a * x, when\ x < 0\ , else\ LReLU(x) = x \ldots \quad (8)$$

The features are augmented via a series of Max Pooling & Drop out operations, which assists in identification of optimal feature sets. These feature sets are classified via a SoftMax based activation layer which is represented as per equation 9,

$$C_{out} = SoftMax\left(\sum_{i=1}^{N} xout_i * W_i + b_i\right) \ldots \quad (9)$$

Where, N represents total number of extracted features by the convolutional layers, while W_i & b_i represents weight matrices which are tuned by the CNN process. The initial network is used to identify image type, which can be either Normal or Malignant, while a similar copy of this network is used to identify different cancer stages. Due to a combination of these layers, the proposed model is able to identify different cancer stages with high-performance levels. These levels are evaluated in the next section of this text, where they are compared with standard cancer identification techniques.

3.5. Feature Fusion

Feature fusion, also known as feature concatenation or feature aggregation, is a crucial technique in deep learning and computer vision for combining information from multiple sources or layers within a neural network. In the context of this work, feature fusion likely plays a significant role in integrating information from different modalities (e.g., frequency, cosine, wavelet, Gabor domains) or phases. It was implemented in the following phases,

i. Motivation for Feature Fusion: Feature fusion is employed to leverage complementary information from various sources or layers to enhance the model's understanding and representation of the data. In this work, it's used to capture important characteristics of breast cancer from different image domains.

ii. Multimodal Feature Extraction: In this work, it was mentioned that histopathological images are converted into multimodal feature sets encompassing frequency, cosine, wavelet, and Gabor domains. Each of these feature sets represents distinct aspects of the image data.

iii. Parallel Processing: Typically, in feature fusion, these multimodal feature sets are processed in parallel. This means that each feature set undergoes its own set of convolutional and pooling operations independently, preserving the unique information from each domain.

iv.Parallel Convolutional Blocks: There may be separate convolutional blocks for each feature set, where each block consists of convolutional layers, batch normalization, activation functions (e.g., ReLU), and pooling layers. These blocks are responsible for capturing relevant features from each domain.

v. Feature Map Dimensions: During the convolutional processing of each feature set, the dimensions of feature maps may change. These dimensions are determined by factors such as filter sizes, padding, and pooling operations. Ensure that the dimensions align properly for subsequent fusion.

vi. Concatenation or Fusion Point: After processing each feature set through its respective convolutional block, the feature maps are concatenated or fused together at a specific point in the network. This point is where the information from different domains is combined.

vii. Fusion Layer: The fusion layer, often implemented as a concatenation layer, combines the feature maps from all domains along a new feature dimension. This results in a unified feature representation that captures information from all modalities.

viii. Further Processing: Following the fusion layer, the unified feature representation may go through additional convolutional and fully connected layers to extract higher-level features and perform classification tasks. These layers can be shared among different domains or specific to the fused features.

ix. Training and Backpropagation: During training, the entire network, including the feature fusion point, was optimized to minimize the chosen loss function. Backpropagation computed gradients to update the model

parameters, ensuring that the fused features contributed effectively to the final classification.

x. Benefits of Feature Fusion: Feature fusion enabled the model to capture the strengths of different feature domains, potentially improving the model's ability to distinguish between benign and malignant tumors. It enhanced model robustness, reduced overfitting, and led to more accurate predictions.

xi. Hyperparameter Tuning: The architecture and hyperparameters for feature fusion, such as the number of convolutional filters, filter sizes, and fusion layer design, were fine-tuned through experimentation to achieve the best results for this work's specific breast cancer detection task.

4. Performance evaluation

The metrics that are used to evaluate the performance of model are accuracy, precision, recall, and F1score. Accuracy indicates the overall correctness of the model's predictions. The degree of precision measures how many of the model's optimistic predictions came true. The model's ability to recognize all relevant instances in the dataset is measured by recall (or sensitivity). High recall means that a significant part of the true positive cases is captured by the model. The harmonic mean of recall and precision makes up the F1-Score, which offers accurate measurement of a model's performance. TP, TN, FP, FN, TPP, and TAP stand for true positive, true negative, false positive, false negative, true anticipated positive, and true actual positive, respectively. The equations used for the computation are shown below. While the TN and FN indicate how many images were correctly or incorrectly classified in a separate class, the TP and FP indicate how many images were correctly and incorrectly classified, respectively.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{TP}{TPP}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{TP}{TAP}$$

$$\text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

5. Results and Discussion

The model is trained on 80% of data consisting of 72000 images and 18000 images are used for model testing. The architecture of the CNN model consists of an Activation layer, convolution layer, batch normalization, second activation and convolution layer and max pooling in multiple phases . The size of input images is defined in terms of 224X224 and in each run, the batch of 64 images were given where weights are updated after every batch of 64 images. The size of the dataset is a critical parameter for deciding the batch size for every run.

The results obtained from model testing are evaluated using different performance evaluation measures including accuracy, recall, precision and F1-score. The results obtained are shown in Table 2. The variation in model accuracy and loss function with different epochs is represented in figure 5. During model training, the accuracy increases and loss reduce smoothly representing the availability of a sufficient number of images to train the model.

Whereas, the considerable fluctuation observed in both accuracy and loss during testing reflects the variation in the number of samples of each class. The confusion matrix reflecting the number of images in a true negative (TN),

true positive (TP), False negative (FN) and false positive (FP) is shown in Figure 6. A large number of images are recorded in TP which defines that images were predicted as true and non-cancerous, and malicious images predicted correctly are recorded as TN. All those images predicted incorrectly are recorded as FP and FN if classified in malicious to benign and benign to malicious respectively.

Table 2: Performance evaluation of model

	Accuracy	Precision	Recall	F1-score
[5]	0.95	0.92	0.94	0.87
[12]	0.91	0.81	0.78	0.86
[16]	0.93	0.86	0.86	0.85
This Work	0.99	0.95	0.98	0.96

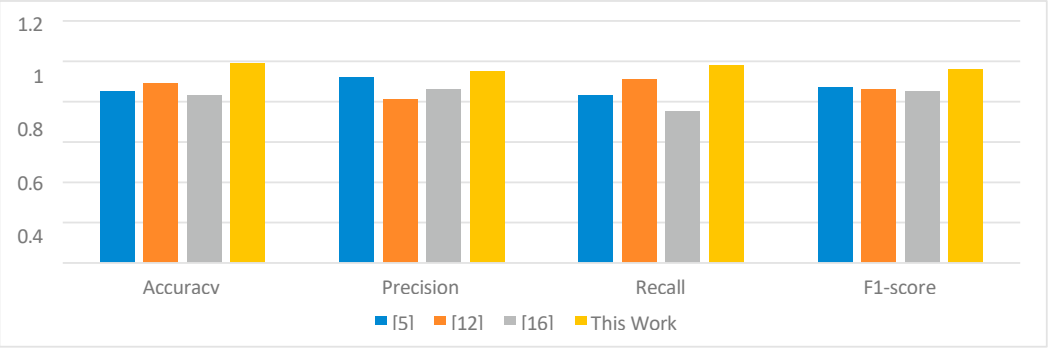


Fig.4. Performance of the MPC2EL model with other models

Performance of the MPC2EL model with other models is shown in the above figure 4.The obtained results reveals that the suggested model exhibits high accuracy, high precision, and high F1 score, making it suitable for usage in a variety of clinical contexts.

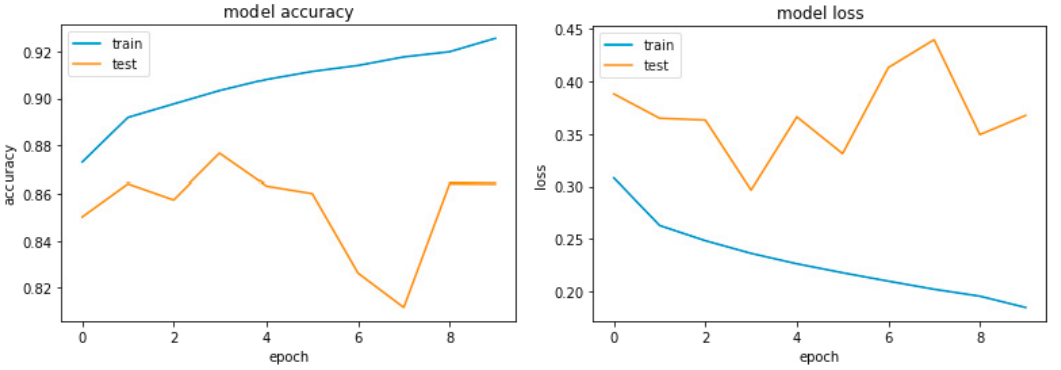


Fig.5. Variation in model accuracy and loss function under different epochs

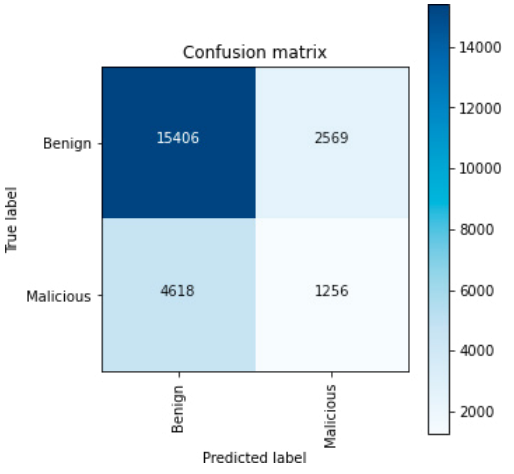


Fig.6. Confusion Matrix plot of benign and malignant classes

Thus, the proposed model achieved 4.5% higher accuracy & precision when compared with [5], it was also able to achieve 3.9% higher recall when compared with [12] & [16] models.

6. Conclusion

This research work has introduced an innovative approach to histopathological image classification, leveraging the power of multi-model fusion and deep learning. Traditional methods of extracting features from histopathological images have been labor-intensive, relying on expert knowledge of breast cancer pathology. Convolutional Neural Networks (CNNs) are used in the suggested method to automatically extract characteristics from a variety of imaging modalities, including MRI, CT scans, ultrasound, and mammogram images, in order to address this problem. The experimental results highlight the superior accuracy achieved by our algorithm, setting a new baseline and demonstrating the potential of deep learning techniques in this domain. In conclusion, the proposed approach not only enhances accuracy but also opens up avenues for future research and refinement, ultimately contributing to more effective and efficient histopathological image analysis in the context of breast cancer.

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